1. INTRODUCTION

1.1 Purpose

This SRS document describes the Odilia's Dispensary Management System (ODMS), version 1.0. The system is designed to automate and streamline the daily operations of St. Odilia's Dispensary, including patient registration, appointment scheduling, medical record management, billing, staff management, and reporting.

The scope of this SRS covers the entire standalone desktop system, developed using Java (JavaFX GUI) and MySQL database integration. It defines the requirements for all six major modules:

- Patient Management
- Appointment Scheduling
- Medical Records
- Staff Management
- Billing and Payments
- Reporting and Analytics

This document serves as a reference point for:

- Developers, to design and implement the system.
- Testers, to validate system functionality against requirements.
- End users (clerks, medical staff, administrators), to understand system capabilities.

Future releases may extend ODMS into a web-based platform with mobile support and integration with national health information systems.

1.2 Intended Audience and Reading Suggestions

This SRS is intended for a diverse group of readers, each with different goals and perspectives:

- **Project Managers:** To understand the scope, objectives, and deliverables of ODMS, ensuring alignment with dispensary goals and healthcare strategies.
- **Developers (Software Engineers):** To use the functional and non-functional requirements as a blueprint for system design, coding, and integration.
- **Testers (Quality Assurance Team):** To validate that the implemented system meets the requirements defined in this document, ensuring reliability and accuracy.
- **Medical Staff (Doctors, Nurses):** To understand how the system supports patient care workflows, including registration, appointments, and medical records.
- **Clerks and Administrative Staff:** To learn how to operate the patient registration, billing, and reporting modules effectively.
- **System Administrators (IT Specialists):** To manage user accounts, roles, backups, and system maintenance.

- **Documentation Writers and Trainers:** To prepare user manuals, training guides, and support materials based on the requirements outlined here.
- **Management and Decision Makers:** To evaluate how ODMS supports operational efficiency, financial transparency, and strategic planning.

Organization of the Document

The SRS is organized into the following sections:

1) **Introduction**: Provides purpose, audience, and scope.
2) **Overall Description:** Explains product perspective, features, user classes, operating environment, constraints, and assumptions.
3) **System Features:** Details functional requirements organized by modules.
4) **External Interface Requirements**: Defines user, hardware, software, and communication interfaces.
5) **Nonfunctional Requirements:** Specifies performance, safety, and security requirements.
6) **Other Requirements:** Covers additional needs such as compliance, upgrades, and database longevity.
7) **Appendices:** Glossary and UML diagrams for clarity.
8) **References:** Lists supporting documents and standards.

Suggested Reading Sequence

- Management and Decision Makers: Begin with Sections 1 and 2 for overview, then proceed to Sections 5 and 6 for nonfunctional and strategic requirements.
- Developers: Focus on Sections 3 and 4 for detailed functional and interface requirements.
- Testers: Read Sections 3 and 5 to design test cases and validation strategies.
- Medical Staff and Clerks: Review Sections 3 and 4 to understand workflows and user interfaces.
- Documentation Writers: Read the entire document, with emphasis on Sections 3, 4, and Appendices for clarity in manuals.

1.3 Project Scope

The Odilia's Dispensary Management System (ODMS) is a healthcare information system designed to automate the core operations of St. Odilia's Dispensary. Its purpose is to replace manual record-keeping and basic spreadsheets with a secure, efficient, and user-friendly digital platform.

Objectives of the System

- **Efficiency**: Reduce paperwork and streamline workflows for patient registration, appointments, billing, and reporting.
- **Accuracy**: Ensure patient data, medical records, and financial transactions are stored and retrieved reliably.

- **Security**: Protect sensitive patient information through role-based access and encryption.
- **Transparency**: Provide clear billing records and management reports for accountability.
- **Scalability**: Design modular features that can be expanded in future releases (e.g., integration with pharmacy or laboratory systems).

Goals

- Deliver a standalone desktop application that supports multiple user roles (Admin, Medical Staff, Clerks, Management).
- Provide interactive graphical interfaces using JavaFX for ease of use.
- Support long-term data storage
- Align with corporate and healthcare strategies, ensuring compliance with Kenya's Ministry of Health ICT standards.

Benefits

- Faster patient service and reduced waiting times.
- Improved accuracy of medical records and billing.
- Enhanced decision-making through analytics and reporting.
- Reduced operational costs by minimizing manual errors and paperwork.

Limitations

- The system will not include pharmacy inventory management or laboratory test integration in version 1.
- The system will not provide mobile application support initially.
- The system will not integrate with external insurance systems in the first release.

2. Overall Description

2.1 Product Perspective

The Odilia's Dispensary Management System (ODMS) is a new, self-contained product designed specifically for St. Odilia's Dispensary. It is a replacement for manual record-keeping methods such as paper files and spreadsheets currently in use.

ODMS is intended to serve as the centralized digital platform for managing dispensary operations. It will integrate multiple modules; Patient Registration, Appointment Scheduling, Medical Records, Staff Management, Billing, and Reporting into a single cohesive system.

System Context:

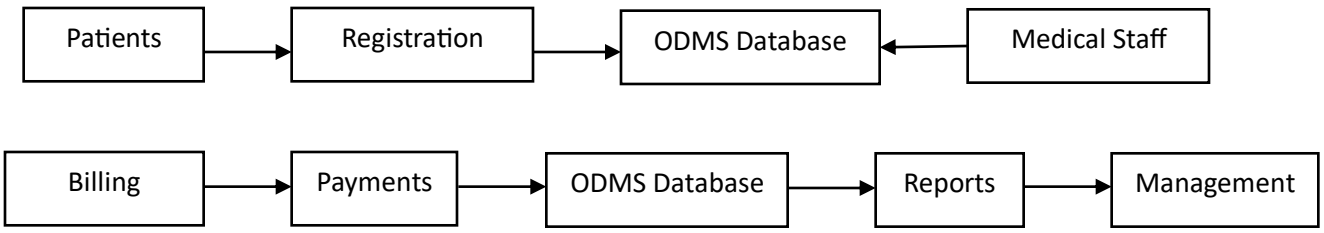
- **Inputs**: Patient demographic details, appointment requests, medical diagnoses, billing information.
- **Processes**: Validation, storage, retrieval, reporting, and analytics.

- **Outputs**: Patient IDs, appointment schedules, medical records, invoices, and management reports.

External Interfaces:

- **Users**: Clerks, medical staff, administrators, and management.
- **Hardware**: Desktop PCs, printers for invoices and reports.
- **Software**: MySQL database, JavaFX GUI.
- **Network**: Local area network (LAN) for multi-user access; cloud backup in future releases.

Diagram – System Context (High-Level):



2.2 Product Features

The Odilia's Dispensary Management System (ODMS) provides a set of integrated modules that collectively automate the dispensary's core operations. At a high level, the system enables the following major functions:

Key Features

Patient Registration and Management

- Capture patient demographic details (name, age, gender, contact information).
- Assign unique patient IDs.
- Update and maintain patient records.

Appointment Scheduling

- Book appointments by patient ID.
- Prevent double-booking through validation.
- Provide calendar views for staff.
- Allow rescheduling and cancellation.

Medical Records Management

- Store patient medical history, diagnoses, and treatments.

- Restrict access based on user roles (doctor, nurse, admin).
- Retrieve patient history quickly for consultations.
- Maintain audit logs of updates.

Staff Management

- Maintain staff profiles (ID, role, privileges).
- Assign roles with specific access rights.

Billing and Payments

- Generate invoices for patient services.
- Record payment status (paid, pending, partial).
- Support multiple payment methods (cash, insurance).

Reporting and Analytics

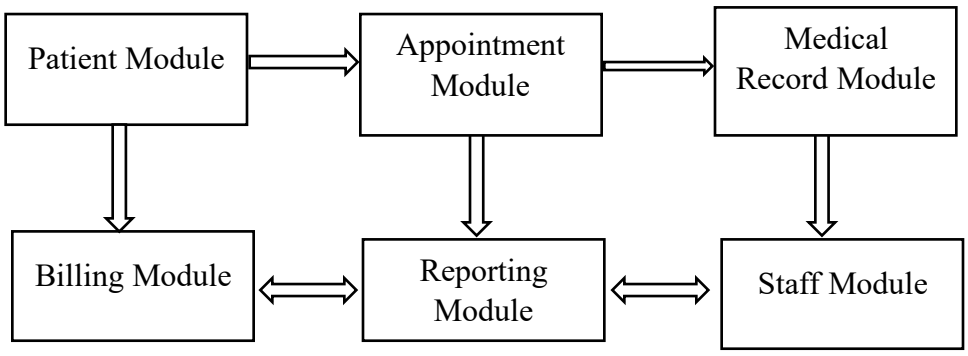
- Generate patient visit reports.
- Produce revenue and financial reports.
- Export reports in PDF/Excel formats.
- Provide filters (by date, staff, service type)

Feature Grouping (Modules Overview)

The features are grouped into six major modules:

- **Patient Module** → Registration, demographic management.
- **Appointment Module** → Scheduling, calendar management.
- **Medical Records Module** → History, diagnoses, treatments.
- **Staff Module** → Roles, privileges, staff activity.
- **Billing Module** → Invoices, payments, financial tracking.
- **Reporting Module** → Analytics, summaries, exports.

Diagram – Top-Level Data Flow (Feature Relationships)



2.3 User Classes and Characteristics

The ODMS will be used by several distinct user classes, each with different responsibilities, levels of expertise, and access privileges. These classes are defined below:

System Administrator (Admin / IT Specialist)

- ***Frequency of Use:*** Daily, for system maintenance and oversight.
- ***Functions Used:*** User account creation, role assignment, backups, security monitoring, system configuration.
- ***Technical Expertise:*** High requires IT knowledge and familiarity with databases and system administration.
- ***Privileges:*** Full access to all modules, including sensitive data and system settings.
- ***Characteristics:*** Responsible for ensuring system availability, data integrity, and compliance with healthcare regulations.

Receptionist / Clerk

- ***Frequency of Use:*** Daily, during patient registration and billing.
- ***Functions Used:*** Register patients, update demographic details, schedule appointments, generate invoices, record payments.
- ***Technical Expertise:*** Basic general office computer skills.
- ***Privileges:*** Access to patient registration, billing, and appointment modules; no access to medical records or system administration.
- ***Characteristics:*** Frontline staff interacting directly with patients, requiring user-friendly interfaces and error prevention mechanisms.

Nurses

- ***Frequency of Use***: Daily, during patient care and follow-up.
- ***Functions Used:*** Access patient records, update medical notes, assist in appointment scheduling, monitor patient progress.
- ***Technical Expertise:*** Moderate basic computer literacy and medical knowledge.
- ***Privileges:*** Access to patient medical records and appointment modules; limited billing access.
- ***Characteristics:*** Require quick and accurate access to patient information to support clinical workflows

Clinical Officer

- ***Frequency of Use:*** Daily, during consultations and treatments.
- ***Functions Used:*** Record diagnoses, prescribe treatments, update patient medical history, review past records.
- ***Technical Expertise:*** Moderate to high medical expertise with basic IT skills.

- ***Privileges:*** Access to medical records, appointment scheduling, and reporting modules; no access to billing or system administration.
- ***Characteristics:*** Primary healthcare providers, requiring secure and reliable access to patient medical data.

Pharmacists

- ***Frequency of Use:*** Daily, when dispensing medication.
- ***Functions Used:*** Review prescriptions, verify patient details, update dispensing records.
- ***Technical Expertise:*** Moderate medical knowledge with basic IT skills.
- ***Privileges:*** Access to medical records (prescriptions) and limited reporting; no access to billing or system administration.
- ***Characteristics:*** Require accurate prescription information to ensure safe and effective medication dispensing.

Laboratory Staff

- ***Frequency of Use:*** Daily, when conducting tests and recording results.
- ***Functions Used:*** Record laboratory test results, update patient medical records, generate lab reports.
- ***Technical Expertise:*** Moderate laboratory expertise with basic IT skills.
- ***Privileges:*** Access to medical records (lab results) and reporting modules; no access to billing or system administration.
- ***Characteristics:*** Require secure and accurate data entry to ensure test results are properly linked to patient records.

Favored User Classes

- Receptionists/Clerks, Nurses, and Clinical Officers are considered primary users, as they interact with the system most frequently and directly affect patient care and operational efficiency.
- Pharmacists and Laboratory Staff are secondary but important users, ensuring continuity of care through prescriptions and test results.
- System Administrators are critical for maintaining system integrity and security.
- Management (covered in reporting) interacts less frequently but relies on accurate data for strategic oversight.

2.4 Operating Environment

The Odilia's Dispensary Management System (ODMS) will operate within a controlled healthcare environment, specifically tailored to the needs of St. Odilia's Dispensary. The environment includes hardware, operating systems, databases, and supporting applications that ensure smooth and secure operation.

Hardware Platform:

- Standard desktop computers with at least: Intel Core i5 or higher, 8 GB RAM, and minimum 500GB HDD or SSD
- Printers for printing invoices, patient cards, and reports
- Thermal Receipt Printer for printing receipts after payments

Network Infrastructure:

- Local Area Network (LAN) for multi-user access

Operating System:

- Windows 10/11 (64-bit editions) and Linux

Database:

- MySQL for patient, staff, billing, and reporting data storage

Software Components

- Development Platform: **Java**
- Graphical User Interface (GUI): **JavaFX** framework for interactive and user-friendly interfaces
- Reporting Tools: Integration with libraries for PDF and Excel export (e.g., Apache POI, JasperReports)
- Backup Tools: Local backup utilities; cloud backup integration planned for future releases

Coexisting Applications

ODMS must coexist with:

- Microsoft Office Suite (Word, Excel): For exporting and reviewing reports
- Antivirus Software: Ensuring system security without interfering with ODMS processes
- Hospital IT Infrastructure: Compatible with existing dispensary hardware and basic networking tools

Environment Characteristics

- ***User Access***: Role-based authentication ensures that each user class (Admin, Receptionist, Nurses, Clinical Officer, Pharmacists, Laboratory Staff) operates within their privilege boundaries.
- ***Data Security:*** The system shall enforce secure access through password-based authentication.
- ***Reliability:*** Designed to operate in environments with occasional power interruptions; backup and recovery mechanisms are included.

2.5 Design and Implementation Constraints

The design and implementation of ODMS is constrained by factors that limit developer options while ensuring compliance, feasibility, and organizational policies.

Corporate and Regulatory Policies

- Must comply with Kenya's Ministry of Health ICT standards and data privacy laws.
- Patient data must be stored securely and handled confidentially.
- Audit trails must be maintained for accountability.

Hardware Limitations

- Must run on standard PCs (Intel Core i5, 4 GB RAM, 500 GB HDD).
- Performance optimized for limited computing resources.
- Must support printers for invoices, patient cards, and reports.

Interfaces to Other Applications

- Must integrate with MySQL databases.
- Reports exportable to Excel and PDF formats.

Specific Technologies, Tools, and Databases

- Language: Java (JDK 17+).
- GUI: JavaFX.
- Database: MySQL or PostgreSQL.
- Reporting: Apache POI or JasperReports.
- Version control: GitHub.

Communications Protocols

- Operates in LAN environment for multi-user access.
- Future versions may support secure cloud backup via HTTPS.
- Data transfer must follow JDBC protocols.

Security Considerations

- Role-based authentication for Admin, Receptionist, Nurses, Clinical Officer, Pharmacists, Lab Staff.
- Sensitive patient data encrypted in database.
- Backup and recovery mechanisms to prevent data loss.

2.7 Assumptions and Dependencies

The development and operation of the Odilia's Dispensary Management System (ODMS) relies on several assumptions and external dependencies. These factors may affect the requirements and implementation of the system.

2.7.1 Assumptions

- **User Competence:** All users (Admin, Receptionist/Clerk, Nurses, Clinical Officers, Pharmacists, Lab Staff) are assumed to have basic computer literacy and can operate the system with minimal training.
- **Hardware Availability:** The dispensary will provide desktop PCs and printers meeting minimum specs (Intel Core i5+, 4 GB RAM, 500 GB HDD).
- **Stable Power Supply:** Assumes reliable electricity with backup (UPS/generators) to handle interruptions.
- **Data Entry Accuracy:** Clerks and medical staff are expected to enter accurate patient and billing information.
- **Regulatory Compliance:** Assumes healthcare regulations and privacy policies remain consistent during the initial release.

2.7.2 Dependencies

- **Database Management System (DBMS):** ODMS depends on MySQL for data storage and retrieval. Changes in DBMS versions or compatibility may affect performance.
- **Third-Party Libraries:** Relies on external libraries (Apache POI, JasperReports) for reporting/export. Updates or licensing changes could impact features.
- **Operating System:** Designed for Windows 10/11. Upgrades to unsupported OS versions may cause compatibility issues.
- **Backup Solutions:** Depends on local backup utilities; future versions may use cloud storage. Reliability of these services is critical for data safety.
- **Future Integration:** May require links to external healthcare systems (e.g., Ministry of Health databases, insurance). Integration needs could affect design and functionality.
- **Maintenance and Support:** Depends on IT staff (Admin) for updates, backups, and troubleshooting. Lack of support could compromise system reliability.

3. SYSTEM FEATURES

3.1 Patient Registration Module

3.1.1 Description and Priority

The Patient Registration Module allows receptionists to register new patients, capture demographic details, and assign unique patient IDs.

Priority: High, because it is essential for system operations.

- Benefit is rated 9 because it ensures accurate patient identification and continuity of care.
- Penalty is rated 8 because without it, patient records would be inconsistent and unreliable.
- Cost is rated 4 because it involves standard data entry forms with moderate development effort.

- Risk is rated 5 because risks include duplicate records or incomplete data entry.

3.1.2 Stimulus/Response (Input/Output)

- Receptionist selects “Register Patient” → system shows form with mandatory fields.
- Receptionist enters details and clicks “Save” → system validates, checks duplicates, assigns ID, and confirms.
- If mandatory fields missing → system shows error: “Please complete all required fields.”
- If duplicate detected → system warns: “Patient already exists.
- On success → patient record becomes instantly available for appointments, billing, and medical records.

3.1.3 Functional Requirements

REQ1: The system shall allow entry of patient demographic details including name, age, gender, contact, and address.

REQ2: The system shall assign a unique patient ID to each registered patient.

REQ3: The system shall validate mandatory fields before saving.

REQ4: The system shall detect duplicate patient records in real time.

REQ5: The system shall display error messages for invalid or incomplete inputs.

REQ6: The system shall confirm successful registration with a message.

REQ7: The system shall make patient records immediately available to other modules.

REQ8: The system shall restrict patient registration functions to authorized staff (Receptionist/Clerk, Admin).

REQ9: The system shall log all patient registration activities in the audit trail.

REQ10: The system shall store patient data securely in the database with encryption for sensitive fields.

3.2 Appointment Scheduling Module

3.2.1 Description and Priority

The Appointment Scheduling Module enables staff to book, reschedule, and cancel patient appointments while preventing double-booking. It ensures smooth patient flow and efficient use of staff time.

Priority: High since it directly impacts service delivery and patient satisfaction.

- Benefit is rated 9 because it improves efficiency and reduces waiting times.
- Penalty is rated 8 because without it, scheduling errors and overlaps would occur.
- Cost is rated 5 because it requires calendar integration and validation logic.
- Risk is rated 6 because risks include incorrect scheduling or overlapping appointments.

3.2.2 Stimulus/Response (Input/Output)

- Receptionist selects “Schedule Appointment” → system shows available slots.
- Receptionist enters patient ID and time → system validates, saves, and confirms.
- If patient already booked → system warns: “Patient already scheduled.”
- If slot unavailable → system shows error: “Selected time is not available.”
- Rescheduling → system updates calendar instantly and confirms.
- Canceling → system removes appointment and logs the action.

3.2.3 Functional Requirements

REQ11: The system shall allow booking of appointments for registered patients.

REQ12: The system shall prevent double-booking of patients or staff.

REQ13: The system shall allow rescheduling and cancellation of appointments.

REQ14: The system shall display available slots within one second after patient ID entry.

REQ15: The system shall confirm successful scheduling, rescheduling, or cancellation.

REQ16: The system shall log all appointment activities in the audit trail.

REQ17: The system shall restrict appointment scheduling functions to authorized staff (Receptionist/Clerk, Admin).

REQ18: The system shall update the calendar instantly when appointments are modified.

REQ19: The system shall display error messages for invalid or unavailable slots.

REQ20: The system shall integrate with patient records to ensure only registered patients can be scheduled.

3.3 Medical Records Management Module

3.3.1 Description and Priority

The Medical Records Management Module provides secure storage and retrieval of patient medical history, including diagnoses, treatments, prescriptions, and laboratory results. It ensures continuity of care and supports clinical decision-making.

Priority: High because it directly affects patient safety and treatment accuracy.

- Benefit is rated 9 because it ensures accurate medical history and continuity of care.
- Penalty is rated 9 because without it, treatment errors and safety risks would arise.
- Cost is rated 6 because it requires encryption, role-based access, and audit logging.
- Risk is rated 7 because risks include unauthorized access or incomplete records.

3.3.2 Stimulus/Response (Input/Output)

- Clinician views records → system shows patient history within 3 seconds.
- Clinician updates records → system validates, saves, and logs action.

- Unauthorized access → system displays “Access denied.”
- Invalid data → system prompts correction and blocks saving.
- Lab results added → system links them to patient record and confirms.
- Updates → synchronized in real time across all modules.

3.3.3 Functional Requirements

REQ21: The system shall store patient medical history securely.

REQ22: The system shall allow authorized staff to view and update medical records.

REQ23: The system shall log all updates, deletions, and access attempts in the audit trail.

REQ24: The system shall restrict medical record access to authorized

REQ25: The system shall retrieve patient history

REQ26: The system shall validate medical record updates before saving.

REQ27: The system shall display error messages for unauthorized access or invalid inputs.

REQ28: The system shall integrate laboratory results into patient records automatically.

REQ29: The system shall synchronize updates in real time across all modules.

3.4 Staff Management Module

3.4.1 Description and Priority

The Staff Management Module allows administrators to create and manage staff profiles, assign roles, and control access privileges. It ensures accountability and security by restricting system functions to authorized personnel.

Priority: Medium because it supports security and operational integrity but does not directly deliver patient care.

- Benefit is rated 7 because it ensures proper access control and accountability.
- Penalty is rated 6 because without it, unauthorized access and misuse of data could occur.
- Cost is rated 5 because it requires role-based access control and privilege management.
- Risk is rated 6 because risks include misconfigured privileges or inactive accounts left open.

3.4.2 Stimulus/Response (Input/Output)

- Admin selects “Add Staff” → system shows form with fields.
- Admin enters details, assigns role, clicks “Save” → system validates, creates profile, confirms.
- Admin updates staff profile → system validates and logs changes.

- Unauthorized access → system shows error: “Access denied. Only administrators can manage staff profiles.”
- Admin deactivates staff account → system removes login privileges and confirms.

3.4.3 Functional Requirements

REQ30: The system shall allow administrators to create staff profiles with personal and professional details.

REQ31: The system shall assign roles with specific privileges (Admin, Receptionist, Nurse, Clinical Officer, Pharmacist, Laboratory Staff).

REQ32: The system shall allow updating of staff details.

REQ33: The system shall allow deactivation of staff accounts to prevent unauthorized access.

REQ34: The system shall restrict staff management functions to administrators only.

REQ35: The system shall validate mandatory fields before saving staff profiles.

REQ36: The system shall log all staff management activities in the audit trail.

REQ37: The system shall enforce password policies for staff accounts.

REQ38: The system shall prevent duplicate staff entries by validating unique identifiers (e.g., staff ID or email).

REQ39: The system shall confirm successful creation, update, or deactivation of staff profiles with a message.

3.5 Billing and Payments Module

3.5.1 Description and Priority

The Billing and Payments Module manages invoices, records payments, and tracks outstanding balances. It ensures financial transparency and sustainability for the dispensary.

Priority: Medium because it supports financial operations but indirectly affects patient care.

- Benefit is rated 8 because it ensures accurate financial tracking and accountability.
- Penalty is rated 7 because without it, billing errors and mismanagement could occur.
- Cost is rated 5 because it requires integration with patient and appointment modules.
- Risk is rated 6 because risks include incorrect billing or incomplete payment records.

3.5.2 Stimulus/Response (Input/Output)

- Receptionist selects “Generate Invoice” → system retrieves details, calculates charges, and displays instantly.
- Payment recorded → system updates invoice status immediately and confirms.

- Invalid payment → system shows error: “Invalid payment input. Please verify details.”
- Partial payment → system records amount, updates balance, and logs transaction.
- Outstanding balances → system generates reminders and includes them in reports.
- All billing and payment activities → logged for accountability and audits.

3.5.3 Functional Requirements

REQ40: The system shall generate invoices automatically based on patient services.

REQ41: The system shall record payment details including method, amount, and date.

REQ42: The system shall update invoice status immediately after payment.

REQ43: The system shall support multiple payment methods (cash, mobile money, insurance).

REQ44: The system shall track outstanding balances and partial payments.

REQ45: The system shall restrict billing and payment functions to authorized staff).

REQ46: The system shall validate payment inputs before saving.

REQ47: The system shall log all billing and payment activities in the audit trail.

REQ48: The system shall confirm successful invoice generation and payment recording with a message.

3.6 Reporting and Analytics Module

3.6.1 Description and Priority

The Reporting and Analytics Module generates patient visit reports, revenue summaries, and staff activity logs. It also provides graphical visualization and export options for management decision-making.

Priority: Medium because it supports administrative oversight rather than direct patient care.

- Benefit is rated 8 because it provides actionable insights and improves operational efficiency.
- Penalty is rated 7 because without it, management would rely on manual records and guesswork.
- Cost is rated 6 because it requires integration with all modules and reporting libraries.
- Risk is rated 5 because risks include inaccurate reporting if data entry is inconsistent.

3.6.2 Stimulus/Response (Input/Output)

- Management selects “Generate Report” → system aggregates data from modules.
- Report generated within 5 seconds → displayed with filters (date, department, staff).
- Graphical analytics requested → charts rendered within 3 seconds.
- Reports → exportable to PDF or Excel.

- Invalid filters → system shows error: “Invalid filter selection. Please adjust criteria.”
- All report activities → logged for accountability.

3.6.3 Functional Requirements

REQ50: The system shall generate patient visit reports with filtering options.

REQ51: The system shall generate revenue and financial reports.

REQ52: The system shall allow exporting reports in PDF and Excel formats.

REQ53: The system shall render graphical analytics within three seconds.

REQ54: The system shall validate filter inputs before generating reports.

REQ55: The system shall restrict reporting functions to authorized roles (Admin, Management).

REQ56: The system shall log all report generation activities in the audit trail.

4. EXTERNAL INTERFACE REQUIREMENTS

4.1 User Interfaces

The ODMS will provide a graphical user interface (GUI) designed with clarity, accessibility, and consistency.

Logical Characteristics:

Login Interface:

- Secure login screen with username and password fields.
- Role-based access control (Admin, Receptionist, Nurse, Clinical Officer, Pharmacist, Laboratory Staff).
- Error messages for invalid login attempts: “Incorrect username or password.”

Dashboard:

- Displays modules (Patients, Appointments, Medical Records, Staff, Billing, Reports).
- Standard navigation menu on the left or top bar.
- Quick access buttons for frequently used functions.

Forms:

- Patient registration, appointment booking, billing, and staff management forms.
- Mandatory fields highlighted with asterisks (*).
- Error messages displayed inline for invalid inputs.

Reports and Analytics:

- Tabular and graphical views (charts, graphs).
- Export options (PDF, Excel).

User Interface Components Needed:

- Login screen
- Patient registration form
- Appointment scheduling calendar
- Medical records viewer/editor
- Staff management form
- Billing and payment form
- Reporting dashboard

4.2 Hardware Interfaces

The ODMS will interact with the following hardware components:

- Desktop Computers: Standard PCs running Windows 10/11 or Linux
- Input via keyboard and mouse.
- Printers: Used to print patient ID cards, invoices, and reports.
- Thermo receipts printer: for printing the receipts after payment
- Input data transmitted via USB interface.
- Network Infrastructure: Local Area Network (LAN) for multi-user access.

Data and Control Interactions:

- Patient data entered via keyboard → stored in database.
- Print commands sent to printer drivers → generate hard copies.
- Barcode scanner input → system retrieves patient record automatically.

4.3 Software Interfaces

The ODMS will integrate with and depend on the following software components:

- Operating System: Windows 10/11 (64-bit).
- Database Management System (DBMS): MySQL for patient, staff, billing, and reporting data.
- Development Platform: Java and JavaFX for GUI development.
- Reporting Libraries: Apache POI for Excel export and JasperReports for PDF generation.
- Office Suite: Microsoft Office (Word, Excel) for report viewing.
- Antivirus Software: Must coexist without interfering with ODMS processes.

Data Sharing Across Components:

- Patient data flows between registration, appointments, medical records, billing, and reporting modules.
- Shared database tables ensure consistency across modules.
- Reports aggregate data from multiple modules.

Implementation Constraints:

- Data sharing must be implemented via secure database transactions.
- No global data areas; modular design ensures separation of concerns.
- Encryption required for sensitive data fields (e.g., patient contact details, medical records).

4.4 Communications Interfaces

The ODMS requires communication functions to support multi-user access, data exchange, and secure interactions between system components.

Requirements:

Network Communication:

- The system shall operate over a Local Area Network (LAN) to enable concurrent access by multiple users.
- Communication between client and server shall use TCP/IP protocols.

Web Browser Access (Future Expansion):

- If migrated to a web-based platform, the system shall support access via modern browsers (Chrome, Edge, Firefox).
- Communication shall be secured using HTTPS with TLS encryption.

Security and Encryption:

- All communication between client and server shall be encrypted using SSL/TLS.
- Sensitive data (e.g., patient records, billing information) shall be transmitted in encrypted form.
- Authentication tokens shall be used to maintain secure sessions.

Data Transfer Rates and Synchronization:

- The system shall support real-time synchronization of patient records, appointments, and billing data across multiple terminals.
- Minimum supported data transfer rate: 100 Mbps LAN.

5. NONFUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

The ODMS must meet defined performance standards to ensure reliability, responsiveness, and scalability under different operating conditions.

a) ***General Performance Requirements:***
   - **System Responsiveness:** The system must respond quickly, with patient registration and appointment actions completed within 2 seconds, and reports generated within 5 seconds for large datasets.
   - **Concurrent Users:** It should support at least 20 users working simultaneously on the LAN without slowing down.
   - **Transaction Throughput:** The system must handle around 100 transactions per minute, covering registrations, updates, and billing.
   - **Scalability:** It should be capable of managing up to 100,000 patient records without requiring major redesign.

b) ***Feature-Specific Performance:***
   - Patient registration validated and saved within 2 seconds, with real-time duplicate detection.
   - Appointment scheduling shows available slots within 1 second and updates instantly when rescheduled or canceled.
   - Medical records retrieved within 3 seconds and updated within 2 seconds.
   - Billing generates invoices within 2 seconds and updates payment status immediately.
   - Reporting produces summaries within 5 seconds and renders analytics within 3 seconds.

c) ***Real-Time Synchronization:***
   - Updates across modules must occur instantly, with conflicts resolved by a "last update wins" mechanism and logged for accountability.

d) ***Rationale:***

   These requirements ensure efficiency, minimize patient waiting times, and provide reliable reporting, guiding developers to optimize database queries, GUI responsiveness, and network communication.

5.2 Safety Requirements

The ODMS must ensure patient safety, data integrity, and operational reliability by preventing loss, damage, or harm that could result from system use.

General Safety Requirements:

Data Backup and Recovery:

- The system shall perform automatic daily backups to prevent data loss.
- Backup files shall be stored securely on local servers and optionally on cloud storage.
- The system shall provide recovery mechanisms to restore data in case of hardware or software failure.

Error Handling:

- The system shall display clear, user-friendly error messages without exposing sensitive technical details.
- Invalid inputs shall be rejected with corrective prompts to prevent incorrect patient or billing records.

Safeguards Against Harm:

- The system shall prevent accidental deletion of patient medical records.
- The system shall log all critical actions (updates, deletions, billing changes) for accountability.
- The system shall restrict access to sensitive modules (medical records, billing) to authorized staff only.

System Failures:

- In case of power outages, the system shall ensure data integrity by saving transactions before shutdown.
- The system shall resume operations without data corruption after restart.

External Policies and Regulations:

- The system shall comply with the Kenya Data Protection Act (2019) regarding patient data handling.
- The system shall align with Ministry of Health guidelines for healthcare information systems.
- The system shall adopt principles from ISO/IEC 62304 (medical device software lifecycle safety) where applicable.

Safety Certifications:

- The system shall meet basic ISO/IEC 27001 standards for information security management.
- The system shall adhere to healthcare IT safety standards to ensure patient data confidentiality and operational reliability.

5.3 Security Requirements

The ODMS must ensure confidentiality, integrity, and availability of patient, staff, and financial data. Security requirements address authentication, privacy, and compliance with healthcare regulations.

Authentication and Access Control:

- The system shall require secure login with username and password.
- The system shall enforce role-based access control (RBAC) to restrict functions to authorized users (Admin, Receptionist, Nurse, Clinical Officer, Pharmacist, Laboratory Staff).
- The system shall lock accounts after 3 failed login attempts and require administrator reset.
- The system shall automatically log out inactive users after 15 minutes of inactivity.

Data Privacy and Protection:

- Patient medical records shall be encrypted in the database using AES-256 encryption.
- Sensitive fields (e.g., contact details, billing information) shall be masked when displayed to unauthorized users.
- The system shall ensure that only authorized staff can view or update medical records.
- Audit logs shall record all access attempts, updates, and deletions.

Communication Security:

- All client-server communication shall be encrypted using SSL/TLS protocols.
- Authentication tokens shall be used to maintain secure sessions.
- Email notifications (if implemented) shall use SMTP with TLS for secure delivery.

Compliance with Regulations:

- The system shall comply with the Kenya Data Protection Act (2019) regarding patient data handling.
- The system shall align with HIPAA principles for healthcare data privacy and security.
- The system shall meet ISO/IEC 27001 standards for information security management.

Security Certifications:

- The system shall satisfy basic ISO/IEC 27001 certification for data security.
- The system shall adopt healthcare IT security best practices to ensure patient confidentiality and operational reliability.

6. OTHER REQUIREMENTS

6.1 Database Requirements

- The system shall use a relational database (MySQL) to store patient, staff, billing, and appointment data.
- The database shall enforce referential integrity between modules (e.g., patient ID linked to appointments, billing, and medical records).
- Sensitive data fields such as patient contact details and medical history shall be encrypted at rest.
- The database shall support backup and restore operations to prevent data loss.

6.2 Internationalization Requirements

- The system shall support English as the default language.
- Date and time formats shall follow regional standards (DD/MM/YYYY, 24-hour **clock).**

6.3 Legal and Regulatory Requirements

- The system shall comply with the Kenya Data Protection Act (2019) for handling patient data.
- The system shall align with Ministry of Health guidelines for healthcare information systems.
- The system shall adopt HIPAA principles for patient privacy and confidentiality.

6.4 Reuse Objectives

- The system shall be designed with modular architecture to allow reuse of components (e.g., reporting engine, authentication module) in future healthcare projects.
- The GUI components shall follow consistent design standards to enable reuse across modules.

6.5 Documentation Requirements

- A separate User Interface Specification shall document detailed screen layouts, workflows, and navigation.
- A System Administration Manual shall describe installation, configuration, and backup procedures.
- A User Manual shall provide step-by-step instructions for staff roles (Receptionist, Nurse, Clinical Officer, Admin).

APPENDIX

Appendix A: Glossary

ODMS: Odilia's Dispensary Management System.

GUI: Graphical User Interface.

SRS: Software Requirements Specification.

RBAC: Role-Based Access Control.

LAN: Local Area Network.

DBMS: Database Management System.

SSL/TLS: Secure Sockets Layer / Transport Layer Security, protocols for secure communication.

HIPAA: Health Insurance Portability and Accountability Act, a US standard for healthcare data privacy.

ISO/IEC 27001: International standard for information security management.

HL7: Health Level Seven, an international standard for healthcare data exchange.

Audit Trail: A secure log of all system activities for accountability.

Backup: A copy of system data stored separately for recovery in case of failure.

Encryption: The process of converting data into a secure format to prevent unauthorized access.

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